

Lecture 01

Introduction to Water Resources Systems Engineering

Course Overview

❖ Water Resources Systems Engineering (수자원시스템공학)

- ◆ 강의 목표: 다양한 수자원시스템에 대해서 현실적인 조건을 만족하는

최적의 정의를 세우고, 그 정의에 따른 최적의 해를 구할 수 있다.

- LP, DP, evolutionary optimization model 등의 여러 가지 최적화 모형의 개념에 대해 이해하고, 이를 실제 문제에 적용할 수 있다.
- 확률론적 모형 (stochastic models)에 대해 이해하고 적용할 수 있다.
- 다양한 평가 지표 (performance criteria)의 특성에 대해 이해하고 이를 적용할 수 있다. (모든 적용은 컴퓨터 언어를 통해 적용, NOT Microsoft Excel)

◆ 강의 키워드 (Keywords of the course)

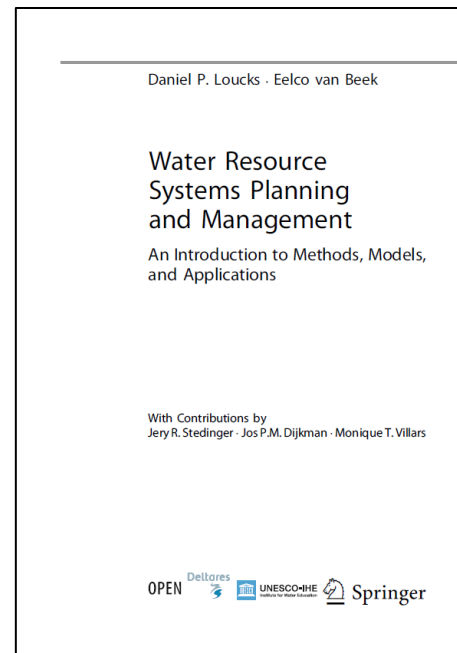
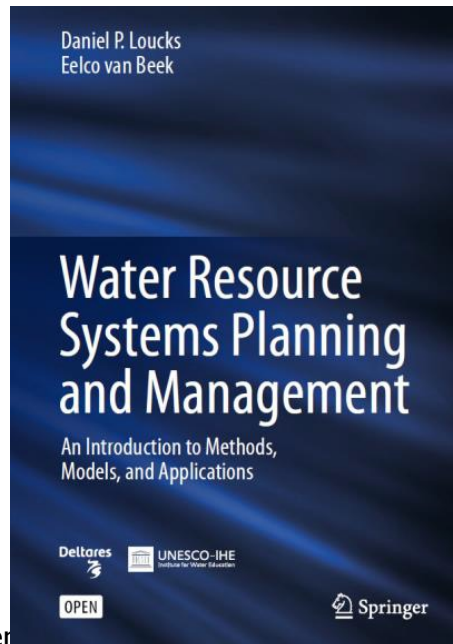
- Part I (week 1-5): system, modeling, traditional optimization models
- Part II (week 6-10): evolutionary models, probability, stochastic, uncertainty

Primary Textbook

- ❖ Loucks, D. P. & Van Beek, E. (2017). **Water Resource Systems Planning and Management: An Introduction to Methods, Models, and Applications**. Springer.

- ◆ <https://ecommons.cornell.edu/handle/1813/48159>

- ◆ **LOOK FOR THE 2017 EDITION, NOT 2005 EDITION**



Course Schedule (Spring 2022)

WEEK/DATE	TOPIC	ASSIGNMENTS QUIZ
1 Mar-07 (M)	<i>Course Introduction</i> Lec 01: Introduction to Water Resources Systems Engineering	
2 Mar-15	Lec 02: Water Resources Systems Modeling <i>Training with Systems Modeling (LT)</i>	
3 Mar-22	Lec 03: Introduction to Conventional Optimization Models (DP & LP) Part I	Assignment 1 Due
4 Mar-29	Lec 04: Introduction to Conventional Optimization Models (DP & LP) Part II	Quiz 1
5 Apr-05	<i>Training with Conventional Optimization Models (LT)</i> <i>Final Project Introduction</i>	
6 Apr-12	Lec 05: Data-fitting, Evolutionary Optimization Models	Assignment 2 Due
7 Apr-19	<i>Training with Evolutionary Optimization Models (LT)</i>	
8 Apr-26	<i>Field Trip (Details TBD)</i>	
9 May-03	Lec 06 Probability, Statistics, Uncertainty, and Stochastic Simulations Part I	Quiz 2
10 May-10	Lec 07: Probability, Statistics, Uncertainty, and Stochastic Simulations Part II	Assignment 3 Due
11 May-17	<i>TEST (Closed Book)</i>	
12 May-24	<i>Training with Stochastic Simulations (LT)</i> Lec 08: Performance Criteria	
13 May-31	Lec 09: Water Resources Systems Engineering Frontiers-Introduction and Examples Part 1	Quiz 3
14 Jun-07	Lec 10: Water Resources Systems Engineering Frontiers-Introduction and Examples Part 2 (<i>Invited Talk</i>)	Assignment 4 Due
15 Jun-14	<i>Final Project Presentation</i> <i>Grading Check</i>	Final Project Report Due

Before Lecture 01

❖ Acronyms (약어)/ Keywords in Lecture 01

- ◆ **WRSA**: Water Resources Systems Analysis (수자원시스템 분석)
- ◆ **WRSE**: Water Resources Systems Engineering (수자원시스템 공학)
- ◆ **IWRM**: Integrated Water Resources Management (통합물관리)
 - “권역 내 물관리에 영향을 미치는 모든 것을 고려하여, 개별적으로 관리 하던 수량·수질·수생태·환경 등을 통합적이고 지능적으로 관리하는 것”
- ◆ **SDGs**: Sustainable Development Goals (지속가능발전목표)
 - 2015년 UN에서 채택된 전세계의 빈곤 문제를 해결하고 지속가능발전을 실현하기 위해 2030년까지 국제사회가 달성해야 할 목표



Overview of WRSA

❖ Problem Statement: Too much, Too little, or Too dirty WATER

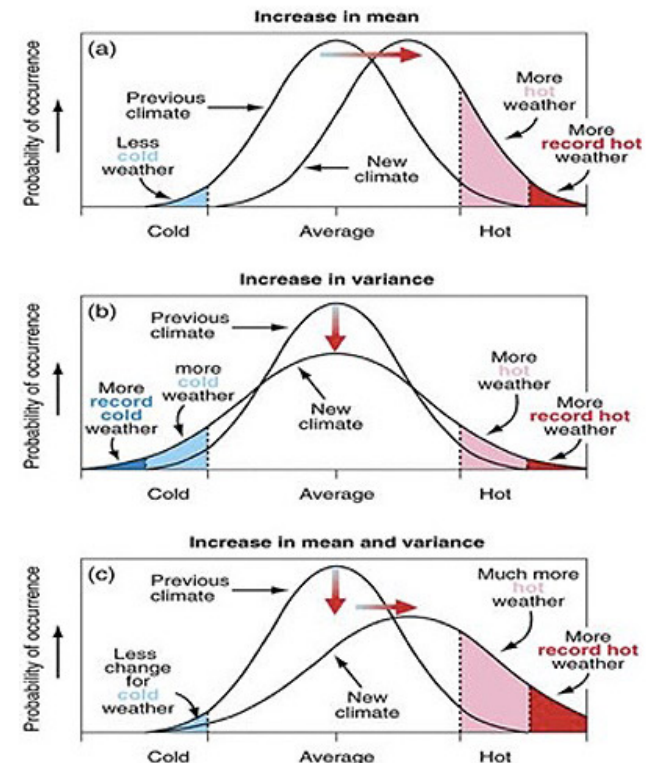
❖ Few Factors Accelerating Dramatic Changes in WRSA

◆ Anthropogenic causes (인위적인 원인)

- Industrialization, human activities affecting droughts, floods, and extreme pollutions in freshwater

◆ (Natural) Hydro-climatic variabilities (자연적인 원인)

- Unexpected outbreaks of floods & droughts from NATURAL variabilities
- 그러나, 인위적 원인과 구분이 어려움



Water Resources Systems Analysis (WRSA)

❖ Range of Water Resources Systems Problems

- ◆ A water resources system is the whole made from connected *hydrologic, infrastructure, ecologic, and human processes that involve water.* (Brown et al. 2015)
- ◆ Essence of WRSA: predicting hydrologic, socioeconomic, & environmental consequences of water management

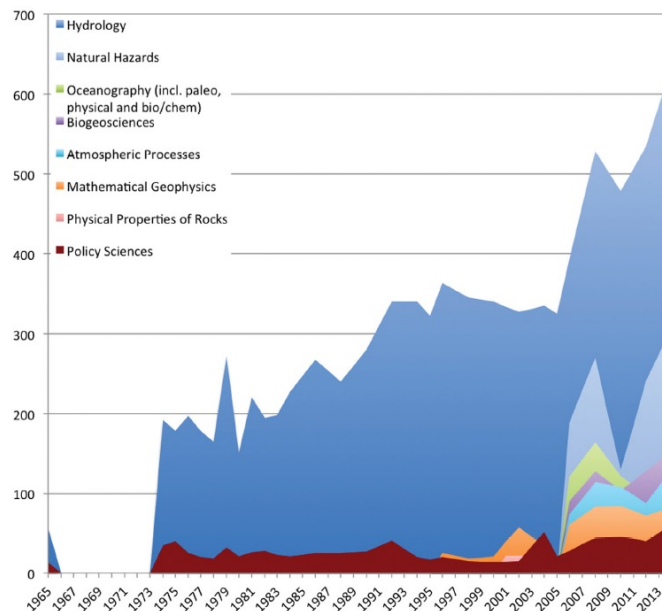


Table 1. Variables for Prediction in Water Resources Systems Analysis

Prediction in Water Resources Systems
Streamflow in basins with human activity present
Groundwater elevation and flow
Reservoir storages and releases
Growth and change in water use in various sectors
Costs and Benefits of water infrastructure investments
Economic value of water
Water-related employment
Agricultural production
Flood damages
Water quality
Ecological status of aquatic and riparian environment
Waterborne disease occurrence

WRSA- “Wicked” Problems?

❖ <D> Wicked: 못된, 사악한

◆ evil or morally bad in principle or practice; [sinful](#); [iniquitous](#):

❖ Water Management = **Wicked** Problems

EDITORIAL

Water Resources Management: The Myth, the Wicked, and the Future

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“The willingness to look ahead, . . . at low-, medium-, and high-level population densities reveals an awareness of the increasing importance of water in the future; and of the necessity for the present generation to plan for the water needs of the next” (Balchin 1960).

Although these issues have been recognized for decades, only a tiny fraction of the water resources research literature blends social, technical, and scientific advances to more directly address the “wicked” nature of water management problems.

3. *The future requires transparency and constructive decision aiding*: Successful examples of consensus-based water management such as “shared-vision modeling” (Lund and Palmer 1997) represent the exception not the rule in the water resources research literature of the past several decades. A review of this *Journal*’s recent literature suggests that problem formulations are most often relegated to brief summaries in our methodologies. Defining our problems to be of use in real decision making is *the problem* in water resources management. Our problem-solving frameworks need to advance

ADDRESSING THE WICKED PROBLEM OF WATER RESOURCE MANAGEMENT: AN ECOSYSTEM SERVICES APPROACH

By E. Hearnshaw, J. Tompkins and R. Cullen
Faculty of Commerce, Lincoln University

WRSA- “*Wicked*” Problems?

❖ Why is water management a *wicked* problem?
(Reed & Kasprzyk, 2009)

1. lack definitive formulations
(the optimal utopia is different for everyone)
2. are not true or false
(judgements can only be subjective not objective)
3. unique and nondecomposable
4. possess decisions that are often irreversible
5. yield a range of consequences that are highly uncertain if not completely unknown

❖ **Future of WRSA?**

Unsolved Problems in Hydrology-UPH

HYDROLOGICAL SCIENCES JOURNAL
2019, VOL. 64, NO. 10, 1141–1158
<https://doi.org/10.1080/02626667.2019.1620507>



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Twenty-three unsolved problems in hydrology (UPH) – a community perspective

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ABSTRACT

This paper is the outcome of a community initiative to identify major unsolved scientific problems in hydrology motivated by a need for stronger harmonisation of research efforts. The procedure involved a public consultation through online media, followed by two workshops through which a large number of potential science questions were collated, prioritised, and synthesised. In spite of the diversity of the participants (230 scientists in total), the process revealed much about community priorities and the state of our science: a preference for continuity in research questions rather than radical departures or redirections from past and current work. Questions remain focused on the process-based understanding of hydrological variability and causality at all space and time scales. Increased attention to environmental change drives a new emphasis on understanding how change propagates across interfaces within the hydrological system and across disciplinary boundaries. In particular, the expansion of the human footprint raises a new set of questions related to human interactions with nature and water cycle feedbacks in the context of complex water management problems. We hope that this reflection and synthesis of the 23 unsolved problems in hydrology will help guide research efforts for some years to come.

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hydrology; science questions; research agenda; interdisciplinary; knowledge gaps

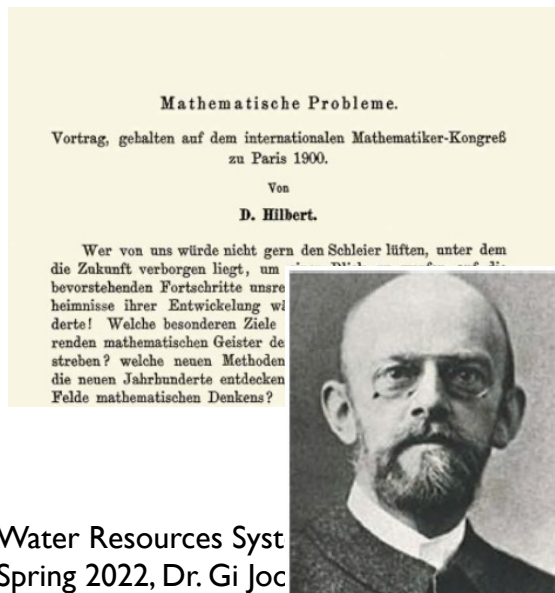
Unsolved Problems in Hydrology-UPH

❖ Background

◆ David Hilbert's 23 Problems in Mathematics (1900)

=> Evolved to Millennium problems (2000)

◆ *"Who among us would not be tempted to lift the veil behind which is hidden the future; to gaze at the coming developments of our science and at the secrets of its development in the centuries to come?"*



List of Unsolved Millennium Prize Problems

- P versus NP.
- Hodge conjecture.
- Riemann hypothesis.
- Yang–Mills existence and mass gap.
- Navier–Stokes existence and smoothness.
- Birch and Swinnerton-Dyer Conjecture.
- ~~Poincaré conjecture.~~

Unsolved Problems in Hydrology-UPH

- ❖ Overall Approach (Blöschl et al. 2019)
 - ◆ **“Why”** questions relating to phenomena
 - ◆ **“What”** questions relating to processes or estimation
 - ◆ **“How”** questions relating to methods



Figure 3. Bottom: Participants of the Symposium on 14 April 2018. Top left: voting in a break-out group. Top right: voting in the plenary session.

Unsolved Problems in Hydrology-UPH

❖ The 23 unsolved problems in hydrology

Table 1. The 23 unsolved problems in hydrology identified by the community process in 2018.

Time variability and change

1. Is the hydrological cycle regionally accelerating/decelerating under climate and environmental change, and are there tipping points (irreversible changes)?
2. How will cold region runoff and groundwater change in a warmer climate (e.g. with glacier melt and permafrost thaw)?
3. What are the mechanisms by which climate change and water use alter ephemeral rivers and groundwater in (semi-) arid regions?
4. What are the impacts of land cover change and soil disturbances on water and energy fluxes at the land surface, and on the resulting groundwater recharge?

Space variability and scaling

5. What causes spatial heterogeneity and homogeneity in runoff, evaporation, subsurface water and material fluxes (carbon and other nutrients, sediments), and in their sensitivity to their controls (e.g. snow fall regime, aridity, reaction coefficients)?
6. What are the hydrologic laws at the catchment scale and how do they change with scale?
7. Why is most flow preferential across multiple scales and how does such behaviour co-evolve with the critical zone?
8. Why do streams respond so quickly to precipitation inputs when storm flow is so old, and what is the transit time distribution of water in the terrestrial water cycle?

Variability of extremes

9. How do flood-rich and drought-rich periods arise, are they changing, and if so why?
10. Why are runoff extremes in some catchments more sensitive to land-use/cover and geomorphic change than in others?
11. Why, how and when do rain-on-snow events produce exceptional runoff?

Interfaces in hydrology

12. What are the processes that control hillslope–riparian–stream–groundwater interactions and when do the compartments connect?
13. What are the processes controlling the fluxes of groundwater across boundaries (e.g. groundwater recharge, inter-catchment fluxes and discharge to oceans)?
14. What factors contribute to the long-term persistence of sources responsible for the degradation of water quality?
15. What are the extent, fate and impact of contaminants of emerging concern and how are microbial pathogens removed or inactivated in the subsurface?

Measurements and data

16. How can we use innovative technologies to measure surface and subsurface properties, states and fluxes at a range of spatial and temporal scales?
17. What is the relative value of traditional hydrological observations vs soft data (qualitative observations from lay persons, data mining etc.), and under what conditions can we substitute space for time?
18. How can we extract information from available data on human and water systems in order to inform the building process of socio-hydrological models and conceptualisations?

Modelling methods

19. How can hydrological models be adapted to be able to extrapolate to changing conditions, including changing vegetation dynamics?
20. How can we disentangle and reduce model structural/parameter/input uncertainty in hydrological prediction?

Interfaces with society

21. How can the (un)certainly in hydrological predictions be communicated to decision makers and the general public?
22. What are the synergies and tradeoffs between societal goals related to water management (e.g. water–environment–energy–food–health)?
23. What is the role of water in migration, urbanisation and the dynamics of human civilisations, and what are the implications for contemporary water management?

Unsolved Problems in Hydrology-UPH

❖ Seven Topics of 23 UPHs

◆ *Time variability and change (시간 변동성과 변화, #1-4)*

- 주로, 기후 변화로 인해 시간적으로 가속화/변화되는 물의 순환에 초점

◆ *Space variability and scaling (공간 가변성 및 범위, #5-8)*

- 공간적 동질성/가변성, 혹은 범위 설정에 따른 문제를 제기

◆ *Variability of extremes (극단에서의 가변성, #9-11)*

- 시공간적 가변성이 아닌, 극치값 (홍수/가뭄)에서의 가변성에 초점
- 기후 변화, 토지 사용 (land-cover/land-use)으로 인해 극심해짐

◆ *Interfaces in hydrology (수문학과와의 접점, #12-15)*

- 위의 주제에 대한 “black box”를 깨는 것
- 어떤 요소로 인해서 시공간 혹은 극단에서의 예상하지 못했던 일이 발생하나?
- 재해/예상하지 못했던 사상의 이유를 과학적으로 규명

Unsolved Problems in Hydrology-UPH

❖ Seven Topics of 23 UPHs

◆ *Measurements and data (측정 및 데이터, #16-18)*

- 리모트 센싱/디지털 데이터 기술을 필두로 한 기술의 발전으로 인해 이전에는 사용할 수 없었던 새로운 정보를 얻을 수 있게 됨
- 시간이 지남에 따라 정보가 쌓이게 되는데, 이를 어떻게 활용할건지?

◆ *Modelling methods (모델링 방법, #19-20)*

- 바뀌는 환경을 반영한 모델링 방법의 변화
(특히 vegetation-식생 변화와 관련해서)
- 모델링 불확실성 (입력자료/매개변수의 불확실성 이외의)에 대한 접근

◆ *Interfaces with society (사회와의 접점, #21-23)*

- 궁극적으로, 이러한 문제를 어떻게 사회에 적용할건지?
- 복잡한 최적화 알고리즘/모델로 구한 해가 전문지식이 상대적으로 낮은 정책결정자들이 선택하기가 어려움.

Assignment #1-1번 문제

❖ Lec 01에서 소개한 23개의 Unsolved Problems in Hydrology (UPH) 중 가장 중요하다고 생각되는, 혹은 가장 시급히 해결되어야 한다고 생각되는 질문은 무엇인가요?

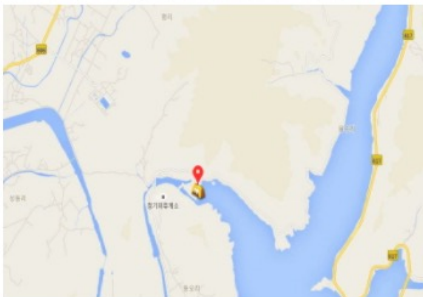
각기 다른 Topic에서 최소 3개의 질문을 선택한 후, 다른 질문들보다 더 중요하다고 생각한 이유(혹은 먼저 해결되어야 한다고 생각하는 이유)를 3개의 질문에 대해 각각 한 문단정도로 적어주세요.

만약 제시된 23개의 문제 이외로 더 중요하다고 생각하는 문제가 있으면 새롭게 제시해 주세요.

Real-world Examples of WRSE in Korea

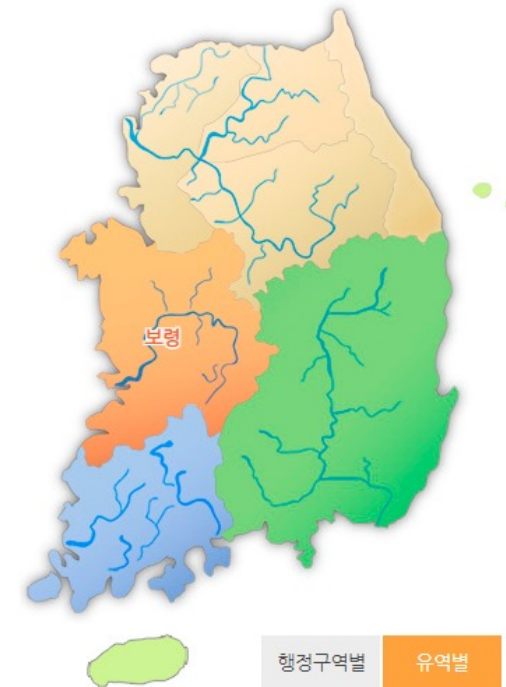
❖ 보령 다목적댐에서의 수자원관리의 어려움

보령댐 현황



자료받기

구분	댐 / 웅천천 / E.C.R.D			
제원	높이 (m)	길이 (m)	정상표고 (EL.m)	체적 (천m ³)
	50.0	291.0	79.0	1,116.0
유역	유역면적 (km ²)		연간용수공급용량 (백만m ³)	
	163.6		106.6	
저수지	저수면적 (km ²)		계획홍수위 (EL.m)	
	5.8		75.50	
	상시만수위 (EL.m)		홍수기제한수위 (EL.m)	
	74.00		74.00	
	월류정표고 (EL.m)		저수위 (EL.m)	
	64.00		50.00	
	총 저수용량 (백만m ³)		유효 저수량 (백만m ³)	
	116.9		108.7	
	홍수 조절용량 (백만m ³)		사업기간	
	10.0		~	



Real-world Examples of WRSE in Korea

❖ 수량의 문제 (TOO LITTLE WATER)

YTN | 2021.08.17. | 네이버뉴스

보령댐 가뭄 경계 단계 진입...금강물 공급

충남 서부지역에 용수를 공급하는 보령댐이 가뭄 경계 단계에 진입하면서 금강물을 공급하는 도수료가 가동에 들어갔습니다. 환경부는 보령댐 저수율이 27% 이하로 떨어져 가뭄 경계 단계에 진입하면서 어제(16일)...

보령댐 가뭄 경계단계 진입, 도수로 가동 용수공급 잡포스트 | 2021.08.17.

보령댐 가뭄 경계단계 진입, 도수로 가동 용... 제주교통복지신문 | 2021.08.17.

보령댐 가뭄 '경계단계' 진입... 도수로 가동 용수... 에너지데일리 | 2021.08.17.

다목적 보령댐 가뭄 경계단계 진입 '비상' 국토일보 | 2021.08.17.

관련뉴스 8건 전체보기 >

충청투데이 | 2015.08.17.

목마른 보령댐... 생활·농업용수도 마른다

이는 보령댐 저수량이 가뭄으로 인해 현재 저수량이 3600만t으로 떨어져 내년 갈수기인 6월까지 급수구역내에 안정적으로 용수를 공급하기 위한 불가피한 조치라는게 관리단의 설명이다. 이와 관련 국토해양부와 K...

대전일보 | 2017.04.28.

극심한 가뭄에 보령댐 저수율 최저 수위 기록

극심한 봄 가뭄이 지속됨에 따라 보령댐의 저수율이 역대 최저치를 기록했다. 27일 한국수자원공사 등에 따르면 이날 보령댐의 저수율은 13%를 기록, 역대 최저치를 기록한 2007년 6월 30일의 13.1%를 갱신했다. 이는...

CMB대전방송 | 2016.08.24.

[CMB뉴스] 계속된 가뭄... 오는 10월 보령댐 '경계' 단계 예상

또한 저수량이 4900만^m³인 보령댐의 현재 저수율은 41.9%이며 이는 예년의 80.4% 수준입니다 국토부 관계자는 이렇게 비가 오지 않고 가뭄이 계속될 경우 오는 10월 중반이면 보령댐은 경계단계로 들어설 가능성이 높...

Real-world Examples of WRSE in Korea

❖ 수질의 문제 (Too Dirty)

연합뉴스 | 2017.06.14. | 네이버뉴스

2급수 금강물 공급받는 보령댐 녹조 '비상'...남조류세포 급증

하천의 수질상태를 확인하는 또다른 기준인 총인(TP)량 역시 2급수인 0.08ppm으로 조사됐다. 금강 백제보~보령댐 도수로 [연합뉴스 자료사진] 금강유역환경청 ...



보령댐 녹조 비상..., 남조류세포수 경계단계까지 치솟아 중도일보 | 2017.06.14.

"용수난 해결도 못했는데"... 보령댐, '녹조' 위기 직면 신아일보 | 2017.06.14.

❖ 경제적 문제 (Too Expensive)

경향신문 | A10면 1단 | 2016.09.18. | 네이버뉴스

625억 들인 보령댐 도수로, 용수 공급 도움 미미...KDI "운영비 매년 90억, 수...

18일 국회 국토교통위원회 더불어민주당 안호영 의원에 따르면 한국개발연구원(KDI) 공공투자관리센터는 최근 펴낸 '보령댐 도수로 건설사업의 비용 부담 방안' 보고서에서 보령댐 도수로 사업의 수익성지수(PI)...

대전일보 | 2015.11.12.

보령댐 도수로 사업비 '줄다리기'

이처럼 해당 사업의 수익규모에 따른 사업비 부담율을 놓고 의견 차이를 빚으면서, 국토부는 KDI에 보령댐 도수로 건설사업에 따른 비용 편익분석을 의뢰했다. 이에 대해 전문가들은 기존 상수도 사업의 경우 운영시...

대전일보 | 2015.12.16.

보령댐 도수로 사업비 의견차 좁힌다

사정이 이렇다 보니 이들은 합리적인 부담 비율 방안을 도출하기 위해 보령댐 도수로 건설에 따른 비용 편익분석을 전문기관에 의뢰하게 된 것. KDI에 의뢰한 해당 용역 결과는 2016년 5월 말까지 나올 것으로 예상돼...

Real-world Examples of WRSE in Korea

❖ Key Questions regarding 보령 다목적댐

- ◆ 문제의 범위를 어떻게 설정할 것인가?
(보령댐 // 연계되어 있는 댐 // 보령댐 의존 행정구역 등)
- ◆ 앞에서 명시된 세 가지 문제 (수량/수질/경제적)에 대한 우선순위는 어떻게 되는가? (가장 우선적으로 해결해야 하는 문제는 무엇인가?)
- ◆ 어떤 방법으로 위의 문제를 해결할 것인가?
(예를 들어, 최적화 모델을 사용해서 // 고위 관계자의 의견을 바탕으로 // 혹은, 관련 모든 이해당사자들-정책결정자, 물 사용자 등- 간의 토의 등)
- ◆ 기타 문제 ...

Conclusion of Lecture 01

❖ Overview of WRSA

- ◆ A water resources system is the whole made from connected ***hydrologic, infrastructure, ecologic, and human processes that involve water.*** (Brown et al. 2015)

❖ WRSA- “***Wicked***” Problems?

- ◆ “**Defining** the problem is the problem itself”

❖ 23 Unsolved Problems in Hydrology (UPH)

- ◆ “Why” “What” & “How” questions
- ◆ Real-world Example of WRSE problem: 보령 다목적댐

❖ **Future of WRSA? (Reed & Kasprzyk 2009)**

- ◆ **Constructive Modeling ... *Interfaces with society***

Future Schedule Preview

❖ 다음 주 강의 (2022-03-15 화요일 17:00-19:50)

◆ Lec 02: Water Resources Systems Modeling

- Modeling Water Resources Systems
- Types of Water Resources Systems Models
- Examples of Water Resources Systems Models

◆ *Training with Systems Modeling (LAPTOP)*

- 과제 #1의 일부분을 수업시간에 진행할 예정이므로, 노트북 지참 (Microsoft Excel, R, Rstudio 미리 설치 필요- 설치 방법 검색)
- 2주 후 2022-03-22 수업 전까지 과제 제출 필요

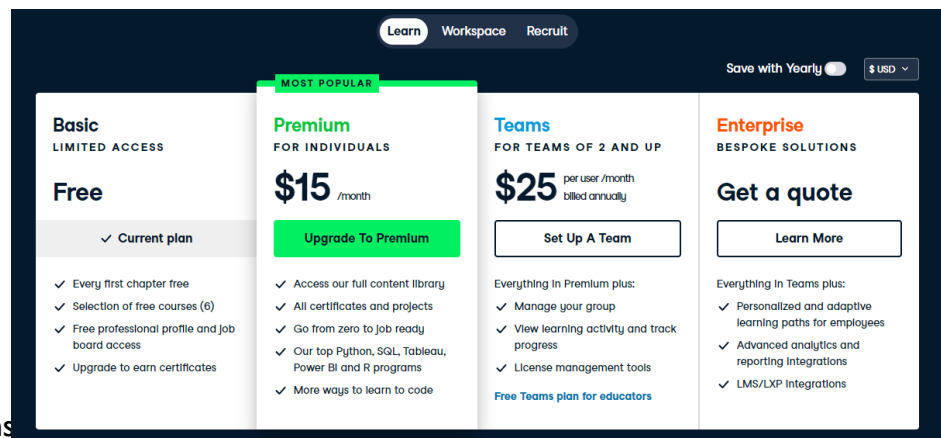
Additional Coding Exercise

PREREQUISITS

- Basic statistics
 - Coding in either of the following computer languages: R, Matlab or Python
- ➔ Credits can be given to students completing additional coding exercise throughout the semester (consult for more details)

www.datacamp.com

- 1) 회원 가입 (Create your free account)
- 2) 일부 콘텐츠 사용 위해서는 Upgrade 필요 (오른쪽 위 보라색 버튼)



Additional Coding Exercise

- ❖ 접속 후: 왼쪽 CATALOG-Tracks-Skill Tracks-R 선택
- ❖ 수행/제출 리스트 (Statement of Accomplishment 제출기한)

- ◆ R Programming (**Mar-29**)

***미리 제출 가능!**

- Introduction to R
- Intermediate R
- Writing Efficient R Code
- Introduction to Writing Functions in R

- ◆ Statistic Fundamentals with R (**Apr-19**)

- Introduction to Statistics in R
- Introduction to Regression in R
- Intermediate Regression in R
- Sampling in R
- Hypothesis Testing in R

Additional Coding Exercise

- ❖ 접속 후: 왼쪽 CATALOG-Tracks-Skill Tracks-R 선택
- ❖ 수행/제출 리스트 (Statement of Accomplishment 제출기한)
 - ◆ Data Visualization with R (**May-31**)
 - Introduction to Data Visualization with ggplot2
 - Intermediate Data Visualization with ggplot2
 - Visualization Best Practices in R
- ❖ 기한에 맞춰서 제출하면 Participation Grading에 반영
 - ◆ 각 제출마다 +3%, 기한 어기거나 미제출시 0%